

Rounak Ghosh

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EDUCATION

Indian Institute of Technology Guwahati

Bachelor of Science in Data Science and Artificial Intelligence

Expected Graduation: Oct 2029 (4 years)

GPA: 7.15

PROJECTS

blas-rs — BLAS Library Implementation in Rust (github)

- Reengineered core BLAS Fortran 77 routines Level 1, 2, and 3 routines from the original Fortran reference in pure Rust while preserving numerical correctness and compatibility with Intel oneMKL/reference BLAS behavior.
- Developed high-throughput computational SIMD kernels by heavily utilizing `std::arch::x86` AVX2 intrinsics (`_mm256_*`) to accelerate vectorized linear algebra operations on x86 processors.
- Optimized instruction-level execution by analyzing emitted assembly to reduce branch mispredictions, enforce memory alignment, and maximize cache-aware data reads.
- Authored an open-source performance engineering handbook documenting compiler behavior, x86 architecture, and native CPU acceleration techniques for numerical computing.

blaze-db — Vector Database Engine (github)

- Architected a high-performance, Rust-native vector database utilizing custom HNSW indexing and SIMD-accelerated metrics, achieving sub-5ms similarity search latency across 400,000+ vector datasets.
- Engineered a highly concurrent read/write architecture using `DashMap` and memory-mapped files, eliminating global lock bottlenecks and handling 800+ concurrent API requests.
- Slashed server-side I/O latency by 99.89x by designing an LRU index caching layer with in-memory checksum validation for immediate cache hits.
- Designed crash-safe storage pipelines using atomic Copy-on-Write (COW), alongside background tombstone-based lazy deletion to optimize disk space.

rdrive — Zero-Trust Distributed Storage Node (github)

- Engineered a multi-tenant, zero-trust object storage node utilizing Content-Addressable Storage (CAS) and Docker-style layered chunking to maximize storage efficiency and deduplication.
- Architected a custom SSH-style handshake protocol utilizing Ed25519 for cryptographic identity verification and X25519 for secure end-to-end encryption.
- Optimized high-concurrency throughput by migrating to a Tokio-based async architecture and implementing per-connection global memory pools to reduce allocation overhead.
- Designed a filesystem-native database utilizing directories as hash tables for fast $O(1)$ lookups and isolated user-space key rotation.

ghost-sync — Async Networking Library for Multiplayer Games (github)

- Built a composable, asynchronous Rust networking library utilizing the Tokio runtime for real-time, non-blocking multiplayer game synchronization.
- Optimized high-frequency broadcast pipelines to handle 195k+ msgs/sec, utilizing zero-copy principles and continuous buffering to sustain high tick-rate performance.
- Managed connection backpressure dynamically, implementing graceful frame-dropping mechanisms to prevent memory exhaustion when client consumption lags behind server dispatch.
- Developed an opinionated, uncoupled API design featuring modular hooks, allowing easy integration for both pure relay and authoritative server architectures.

TECHNICAL SKILLS

Languages: Rust, C, Java, Bash, Perl

Technologies: Linux, Docker, Tokio, Axum, Intel VTune, Oracle Cloud, WireShark

Interests: Systems and Network Programming, Compilers, Databases, CLI Tools, Game Dev, Performance engineering